Modeling Laser Source Term in an Reaction Flow Solver

Loc Duong, Brian Kiedrowski

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When modelling laser-droplet interactions, we propose adding a volumetric source term q''' into the energy conservation equation of the exisiting solver described in [1].

$$\frac{\partial \rho E}{\partial t} + \nabla \cdot [\rho \mathbf{u} H] = \nabla \cdot \alpha \nabla T + \nabla \cdot [\boldsymbol{\tau} \cdot \mathbf{u}] + \sum_{k=1}^{N_s} h_k \nabla \cdot \rho D \nabla Y_k + q'''. \tag{1}$$

As is finite volume formulation, Equation 1 has to be integrated over a control volume, which is a cell on the mesh. This turns q''' into q_i, the total heat generation rate within cell i.

$$\iiint_{V_i} q''' dV = q_i = q_{0,i}(1 - e^{-\mu s_i}), \tag{2a}$$

where

$$q_{0,i} : \text{laser energy upon entering cell } i \tag{2b}$$

$$\mu : \text{attenuation coefficient of the medium} \tag{2c}$$

$$s_i : \text{traveled distance in cell } i. \tag{2d}$$

It is expected that the laser heat source vector $\mathbf{q}$ is a pretty sparse vector, as the majority of cells are not in the laser path. In such cells, $s_i = 0$ and $q_i = 0$. Computing the laser heat source is a ray-tracing problem to identify the few cells that intersect the laser.

We tested the heat source term on a structured ellipsoidal mesh, where the basis variables r, μ, and φ are defined as followed:

$$\mathbf{r}(r, \mu, \varphi) = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} ar\mu \\ br\sqrt{1-\mu^2}\cos\varphi \\ cr\sqrt{1-\mu^2}\sin\varphi \end{bmatrix}, \tag{3}$$

where $r \in [0, 1]$, $\mu \in [-1, 1]$, and $\varphi \in (-\pi, \pi]$ are the coordinate variables. The axes x', y', and z' form a rotated Cartesian grid where x' points to the semi-major direction, y' to the semi-middle direction, and z' to the semi-minor direction. As such, the polar cosine μ is always measured from the semi-major axis. Note that if $a = b = c$, the coordinate system is spherical and orthogonal.

Each cell contains up to six surfaces, each of which corresponds to a basis variable being held constant either at its lower or upper bound. The ray-tracing algorithm will calculates the distance that a ray entering a cell will travel to arrive at each of the six surfaces, and the smallest positive value is assigned to s_i. Denote $\mathbf{r}_i$ and $\hat{\boldsymbol{\omega}}_i$ as the entrance position and direction of the ray, respectively, in cell i, our goal is solve for the distance travelled s which that $\mathbf{r}_i + \hat{\boldsymbol{\omega}}_i s$ is on the boundary of cell i.

Surfaces of constant r are ellipsoids, and the distance to any such surface

$$(\mathbf{r}_i + \hat{\boldsymbol{\omega}}_i s) \cdot \mathbf{M}_r (\mathbf{r}_i + \hat{\boldsymbol{\omega}}_i s) = 1 \tag{4a}$$

$$\mathbf{M}_r \equiv \operatorname{diag}\begin{bmatrix} \frac{1}{a^2} & \frac{1}{b^2} & \frac{1}{c^2} \end{bmatrix}. \tag{4b}$$

Equation 4 is equivalent to the quadratic equation

$$As^2 + 2Bs + C = 0, \tag{5a}$$

where

$$A = \hat{\boldsymbol{\omega}}_i \cdot \mathbf{M}_r \hat{\boldsymbol{\omega}}_i \tag{5b}$$
$$B = \mathbf{r}_i \cdot \mathbf{M}_r \hat{\boldsymbol{\omega}}_i \tag{5c}$$
$$C = \mathbf{r}_i \cdot \mathbf{M}_r \mathbf{r}_i - 1 \tag{5d}$$

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Surfaces of constant μ are elliptic cones, and the distance to any such surface satisfies

$$(\mathbf{r}_i + \hat{\boldsymbol{\omega}}_i s) \cdot \mathbf{M}_\mu (\mathbf{r}_i + \hat{\boldsymbol{\omega}}_i s) = 0 \tag{6a}$$
$$\mathbf{M}_\mu \equiv \operatorname{diag}\begin{bmatrix} \frac{1-\mu^2}{a^2} & -\frac{\mu^2}{b^2} & -\frac{\mu^2}{c^2} \end{bmatrix}. \tag{6b}$$

It's worth noting that Equation 6 include surfaces μ and $-\mu$, so it is important to ensure that $\mathbf{r} + \hat{\boldsymbol{\omega}}\mathbf{s}$ lands on the correct surface.

Surfaces of constant φ are ellipses, and the distance to any such surface satisfies

$$\frac{y_i' + \omega_{y',i} s}{b} \sin\varphi = \frac{z_i' + \omega_{z',i} s}{c} \cos\varphi, \tag{7}$$

which is only a linear equation in s. After solving for s it is important to ensure the correct surface is landed on, as Equation 7 is identical for φ and $\varphi \pm \pi$.

A major assumption built in the ray-tracing algorithm is that the path a ray takes within each cell is a straight line, and the cell refractive index is calculated from cell-averaged quantities, i.e. $n_i = n(\rho_i, T_i)$. Snell's Law is used to relate the exit direction in cell i to the entrance direction in cell j. Once the distance travelled in cell i is solved for, the updated variables are as followed:

$$q_{0,j} = q_{0,i} e^{-\mu s_i} \tag{8a}$$
$$\mathbf{r}_j = \mathbf{r}_i + \hat{\boldsymbol{\omega}}_i s_i \tag{8b}$$
$$n_j (\hat{\mathbf{n}}_i \times \hat{\boldsymbol{\omega}}_j) = n_i (\hat{\mathbf{n}}_i \times \hat{\boldsymbol{\omega}}_i) \tag{8c}$$

Here, $\hat{\mathbf{n}}_i$ is the normal vector on the surface that the ray exits from cell i.

To demonstrate the laser source term, a solver is built to solve the simplified energy conservation equation

$$\frac{\partial \rho E}{\partial t} = \nabla \cdot \alpha \nabla T + q'''. \tag{9}$$

All dependencies on velocity and species terms are neglected, and the droplet is treated as if it is a single-species low-density block of solid. The diffusive flux is approximated using central difference, and the time-integration scheme is the Explicit Midpoint Method. The initial laser heat generation rate is set to 1000

W. Physical properties are taken from water properties, and thermal expansion is assumed to be uniform.

Figure shows the temperature distribution of a hypothetical droplet at $t = 7.5$ s. The cells surrounding the laser trajectory show a much higher temperature than the surrounding ones due to a lack of effective heat transfer modes.

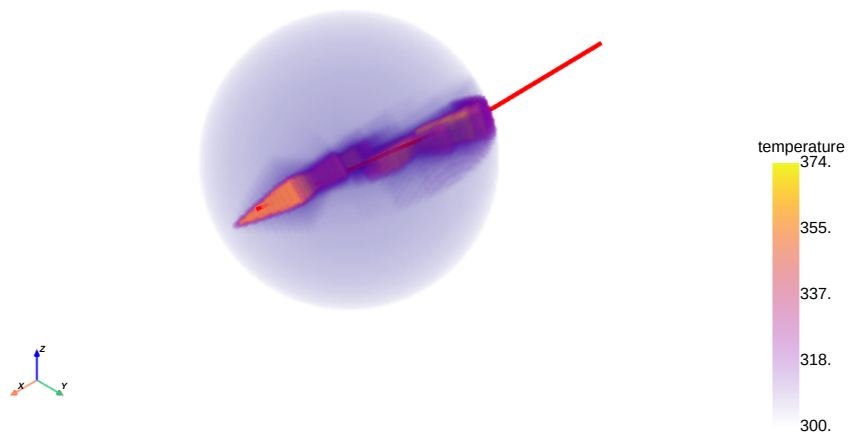

Figure 1: Temperature distribution of a droplet undergoing laser interaction, solved using Equation 9. The red line is the laser trajectory.

Bibliography

[1] S. Sharma et al. *An AMReX-based Compressible Reacting Flow Solver for High-speed Reacting Flows relevant to Hypersonic Propulsion*. 2025.